ACADEMIC RESEARCH PROJECT (MBA-WILP)

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE PILANI

EC1: 3 Phase Mini Project Plan

Applied Data Science (S1-25_MBACCZG581)

PHASE 1

Housing Price Prediction – Boston Housing dataset (UCI ML).

Submitted by: **Group 11**

Sno	Name	BITS ID
1	Akhilesh Pandey	2024bc26663@wilp.bits-pilani.ac.in
2	Awanish Kumar	2024BC26664@wilp.bits-pilani.ac.in
3	Narmatha S	2024BC26642@wilp.bits-pilani.ac.in
4	Manoj Kesanupalli	2024BC26635@wilp.bits-pilani.ac.in

Submission Date: 24-Dec- 2025

Executive Summary

This project focuses on exploring the Boston Housing dataset to understand the key factors that influence housing prices. The analysis aims to provide insights that can guide real estate strategies and urban planning decisions. Rather than building predictive models, the emphasis is on identifying patterns and relationships through exploratory data analysis. This approach will help stakeholders make informed decisions about housing development, affordability, and community planning. Scope Clarification: This Phase-1 report focuses only on defining the problem, understanding the dataset structure, feasibility, and initial exploratory notes. No data cleaning, no visualizations, and no modelling steps are included at this stage as per EC1 Phase-1 requirements.

Context

Housing prices are influenced by multiple factors such as crime rates, number of rooms, accessibility to highways, and school quality. Understanding these relationships is critical for real estate companies, policymakers, and urban planners. The Boston Housing dataset provides a rich set of variables that reflect socio-economic and environmental aspects of neighborhoods. This project uses this dataset as a case study to demonstrate how data-driven insights can support better decision-making.

Acknowledgement

We sincerely thank BITS Pilani Faculty (Shree Gokhale Jayanti) for providing this opportunity to apply data science concepts to real-world problems. We also acknowledge the support of our peers and the availability of public datasets from UCI Machine Learning Repository and Kaggle, which made this analysis possible.

Contents

1. Problem statement 5

2. Dataset description & access methods 5

3. Exploratory notes 6

4. Any other aspects 9

5. Academic Declarations 10

Abbreviation Table

Abbreviation	Full Form
CRIM	Crime rate per capita
ZN	Residential land zoned for large lots (%)
INDUS	Non-retail business acres (%)
CHAS	Charles River dummy variable
NOX	Nitric oxide concentration
RM	Average number of rooms per dwelling
AGE	% of houses built before 1940
DIS	Distance to employment centers
RAD	Index of highway accessibility
TAX	Property tax rate
PTRATIO	Pupil-teacher ratio
B	Race-based calculation
LSTAT	% of lower-status population
MEDV	Median house value ($1000s)

1. Problem statement

- We want to understand what factors influence house prices in Boston and the goal of this project is to understand those factors influence house prices in Boston.
- Real estate companies and policymakers can use this information to make better decisions about property investments, urban planning, and housing affordability.
- This will help real estate companies and policymakers identify key drivers like number of rooms, crime rate, and school quality.
- Instead of building predictive models, we will focus on exploring the data to identify patterns and relationships between house prices and other factors such as crime rate, number of rooms, and school quality.

2. Dataset description & access methods

- **Source:** Boston Housing dataset (public dataset from UCI Machine Learning Repository, downloaded from Kaggle). https://archive.ics.uci.edu/dataset/2/boston+housing.
- **File Format:** Space-separated text file (.csv) without header row.
- **Size:** 506 rows and 14 columns.
- **Access Method:** Downloaded from Kaggle and opened in Notepad and Excel using "space" as delimiter.
- **Columns and Meaning:**
  1. CRIM – Crime rate per capita.
  2. ZN – Percentage of residential land zoned for large lots.
  3. INDUS – Percentage of non-retail business acres.
  4. CHAS – Charles River dummy (1 = near river, 0 = not).
  5. NOX – Nitric oxide concentration (air pollution).
  6. RM – Average number of rooms per house.
  7. AGE – Percentage of houses built before 1940.
  8. DIS – Distance to employment centers.
  9. RAD – Index of highway accessibility.
  10. TAX – Property tax rate.
  11. PTRATIO – Pupil-teacher ratio in schools.
  12. B – A calculated value related to race.
  13. LSTAT – Percentage of lower-status population.
  14. MEDV – Median house value (in $1000s).

3. Exploratory notes

- The dataset has no header row, so column names must be added manually.
- No missing values were found in the dataset.
- Ranges observed:
    - House Price (MEDV) ranges from 5.0 to 50.0 (values capped at 50).
    - Rooms (RM) range from 3.5 to 8.7.
    - Crime Rate (CRIM) has very large values for some areas, indicating possible outliers.
- Columns CHAS (0 or 1) and RAD (index values) are categorical.
- Some columns like CRIM and LSTAT show skewed distributions, which will need attention during visualization.

- 0.00632 → CRIM
- 18.00 → ZN
- 2.310 → INDUS
- 0 → CHAS
- 0.5380 → NOX
- 6.5750 → RM
- 65.20 → AGE
- 4.0900 → DIS
- 1 → RAD
- 296.0 → TAX
- 15.30 → PTRATIO
- 396.90 → B
- 4.98 → LSTAT
- 24.00 → MEDV (Price)

"Column names and their descriptions are sourced from the official UCI Machine Learning Repository documentation for the Boston Housing dataset. The raw Kaggle file does not include headers; these were added based on UCI's published schema."

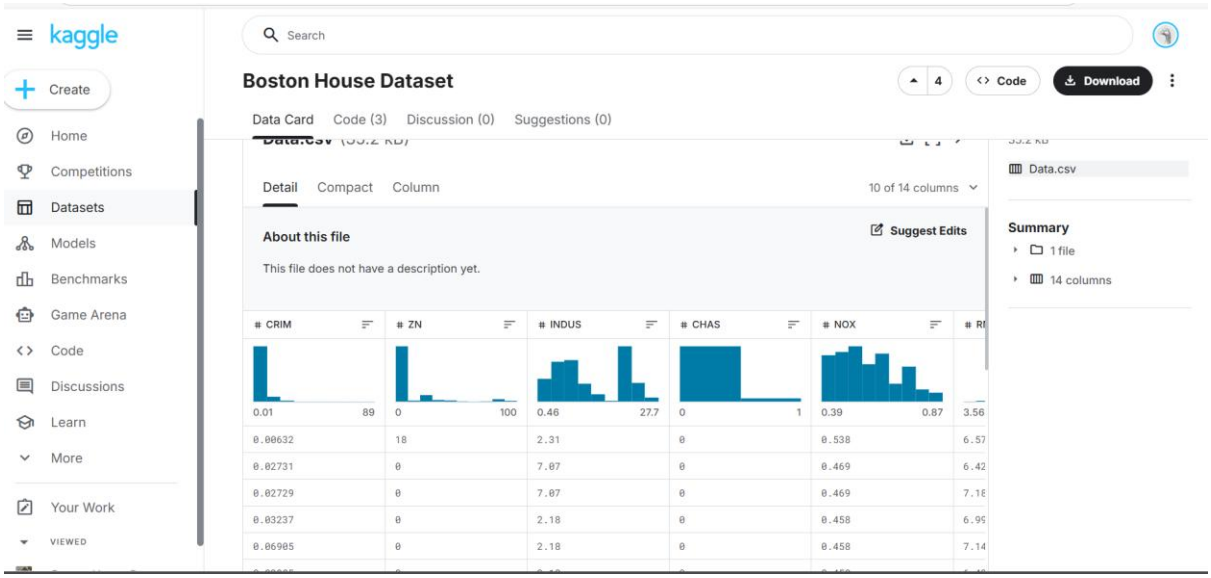
3.1 Feasibility of the Dataset

- The Boston Housing dataset is suitable for this project because it offers a wide range of variables representing environmental, social, and structural characteristics of neighbourhoods
- These factors are directly related to real estate pricing and are commonly analyzed in urban economics.
- The dataset has a manageable size (506 rows × 14 columns), contains no missing values, and all variables are numerical except one categorical field (CHAS).
- This makes preprocessing and exploratory analysis straightforward for beginners while still offering meaningful insights.
- Some challenges include skewed distributions, outliers in variables such as CRIM and TAX, and capped price values at 50.
- These issues provide good learning opportunities for data cleaning and transformation in Phase 2.

3.2 Phase-1 Data Limitations:

- MEDV values are historically capped at 50.0 (top-coded).
- Dataset is older and may not reflect current Boston demographics.
- Raw file contains no headers; requires mapping from UCI documentation.
- These exploratory notes confirm that the dataset is clean (no missing values), numerical, and suitable for Phase-2 EDA.
- Early patterns such as skewed CRIM and LSTAT distributions indicate the need for careful visualization techniques in the next phase.

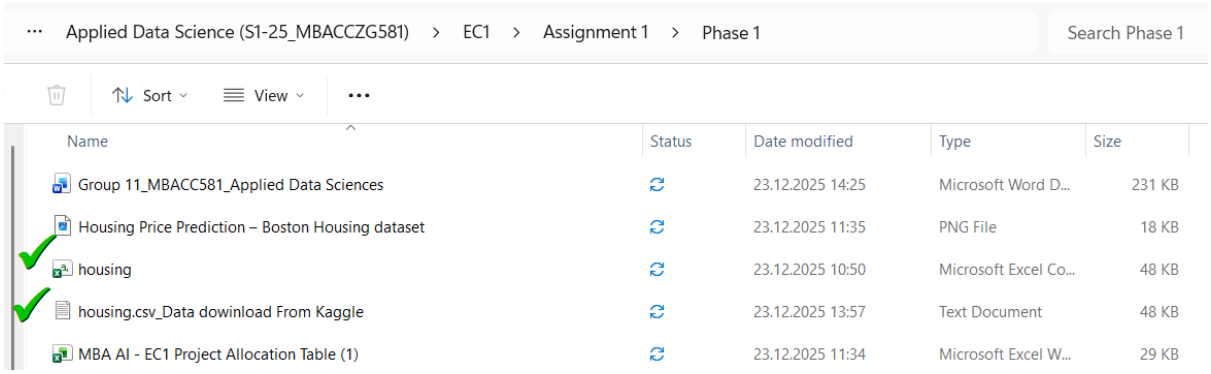
Data downloaded from Kaggle



Downloaded data open in notepad file

```
0.00632  18.00   2.310  0  0.5380  6.5750  65.20  4.0900   1  296.0  15.30 396.90   4.98  24.00
0.02731   0.00   7.070  0  0.4690  6.4210  78.90  4.9671   2  242.0  17.80 396.90   9.14  21.60
0.02729   0.00   7.070  0  0.4690  7.1850  61.10  4.9671   2  242.0  17.80 392.83   4.03  34.70
0.03237   0.00   2.180  0  0.4580  6.9980  45.80  6.0622   3  222.0  18.70 394.63   2.94  33.40
0.06905   0.00   2.180  0  0.4580  7.1470  54.20  6.0622   3  222.0  18.70 396.90   5.33  36.20
0.02985   0.00   2.180  0  0.4580  6.4300  58.70  6.0622   3  222.0  18.70 394.12   5.21  28.70
0.08829  12.50   7.870  0  0.5240  6.0120  66.60  5.5605   5  311.0  15.20 395.60  12.43  22.90
0.14455  12.50   7.870  0  0.5240  6.1720  96.10  5.9505   5  311.0  15.20 396.90  19.15  27.10
0.21124  12.50   7.870  0  0.5240  5.6310 100.00  6.0821   5  311.0  15.20 386.63  29.93  16.50
0.17004  12.50   7.870  0  0.5240  6.0040  85.90  6.5921   5  311.0  15.20 386.71  17.10  18.90
0.22489  12.50   7.870  0  0.5240  6.3770  94.30  6.3467   5  311.0  15.20 392.52  20.45  15.00
0.11747  12.50   7.870  0  0.5240  6.0090  82.90  6.2267   5  311.0  15.20 396.90  13.27  18.90
0.09378  12.50   7.870  0  0.5240  5.8890  39.00  5.4509   5  311.0  15.20 390.50  15.71  21.70
0.62976   0.00   8.140  0  0.5380  5.9490  61.80  4.7075   4  307.0  21.00 396.90   8.26  20.40
```

Data repository -



4. Any other aspects

Plan for Next Phase (Phase-2):

- Add headers to the dataset.
- Perform data cleaning and handle outliers.
- Create initial charts:
    - Price vs Rooms
    - Price vs Crime Rate
    - Price vs PTRATIO
- Summarize findings with business meaning.

Ethical Considerations for Phase-2:

- Data is from a public source.
- Ensure fair interpretation without bias or misuse.

_____Report Ends Here_____

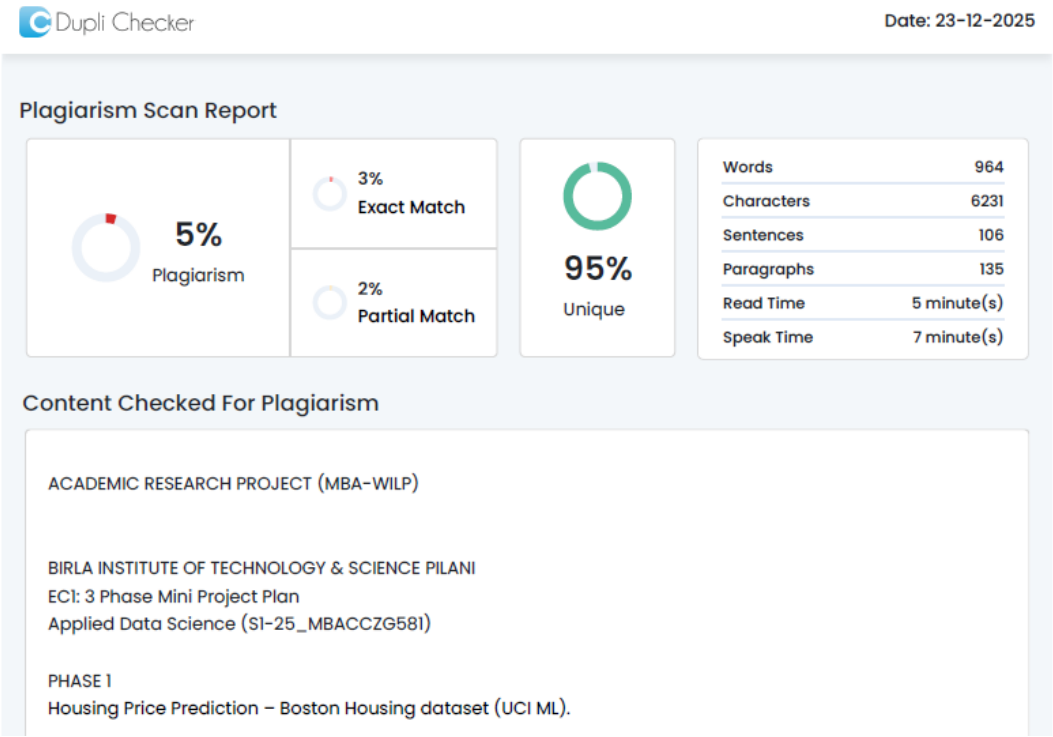
5. Academic Declarations

References

- Raw Boston Housing dataset was sourced from:
- - UCI ML Repository (Harrison & Rubinfeld, 1978) - Kaggle (public dataset) Dataset is space-separated and originally contains no header row.
- Kaggle. (n.d.). Boston Housing Dataset. Retrieved from https://www.kaggle.com/datasets

Plagiarism Declaration

"I hereby declare that this assignment is my original work and has not been copied from any other source. All references and supporting materials have been appropriately acknowledged."



Intellectual Property & Copyright Notice

"This document is prepared solely for academic purposes as part of the AI Application Ecosystem course. All proprietary information related to TATA Technologies GmbH is masked or generalized to maintain confidentiality. Any reproduction or distribution of this content without permission is prohibited."

AI Tools Usage Declaration

"AI tools were used to assist in structuring ideas, improving clarity, and generating examples. The final content was reviewed, edited, and validated by me to ensure accuracy and relevance to my real work context."

****